

Assignment: Understanding Database Management Systems

Objective:

The objective of this assignment is to provide hands-on experience in designing and implementing a database system. Students will create ER diagrams, normalize data, convert ER diagrams to relational schemas, and write SQL queries to interact with the database.

Case Study: University Enrollment System

Scenario:

You are tasked with creating a database to manage student enrollments in classes and students' final grades. The university is organized into colleges, and each college administers a number of academic departments. Departments offer various courses, and courses are delivered in sections. Students enroll in sections and receive grades for their performance.

Requirements:

Part A

Question 1. ER Diagram:

Design an ER diagram based on the following requirements:

- The university is organized into colleges (COLLEGE).
Attributes: CName (unique), COffice, CPhone
Each college has a dean (Faculty member).
- Each college administers several departments (DEPT).
Attributes: DName (unique), DCode (unique), DOffice, DPhone, CStartDate (when the current chair started).
Each department has a chair (Faculty member).
- Each department offers several courses (COURSE).
Attributes: CoName (unique), CCode (unique), Level, Credits, CDesc
- The database keeps track of instructors (INSTRUCTOR).
Attributes: Id (unique), IName, IOffice, IPhone, Rank
Each instructor belongs to one primary department.
- The database stores student information (STUDENT).
Attributes: Sid (unique), FName, MName, LName, Addr, Phone, Major, DoB
Each student belongs to one primary department.

- Courses are offered as sections (SECTION).
Attributes: SecId (unique), SecNo, Sem, Year, CRoom, DaysTime
Each section is associated with one course and one instructor.
- The database keeps track of enrollments (ENROLLMENT).
Attributes: Sid, SecId, Grade

Question 2. Normalization:

Normalize the database to the Third Normal Form (3NF). Ensure that each relation is free of update, insertion, and deletion anomalies.

Question 3. Relational Schemas:

Convert the ER diagram into relational schemas. Define primary keys, foreign keys, and other constraints as necessary.

Part B

Question 4. SQL Queries:

Write SQL queries to accomplish the following tasks:

1. List all colleges with their deans.
2. Find all departments within a specific college.
3. Retrieve the details of the faculty member chairing a specific department.
4. List all courses offered by a specific department.
5. Find all sections of a specific course offered in a particular semester and year.
6. List all students enrolled in a specific section.
7. Calculate the average grade for a student across all sections they have completed.
8. Find all instructors teaching a specific level of course (e.g., junior level).

Instructions:

1. ER Diagram:

- Use any ER diagram drawing tool (e.g., Lucidchart, draw.io) to create your ER diagram.
- Ensure that all entities, attributes, and relationships are correctly represented.

2. Normalization:

- Identify functional dependencies in the initial schema.
- Decompose relations to eliminate partial and transitive dependencies.
- Provide a step-by-step explanation of how you achieve 3NF.

3. Relational Schemas:

- Write the relational schemas in the form:
- `TableName(Attribute1, Attribute2, ..., AttributeN)`
- Identify primary keys and foreign keys in each table.

4. SQL Queries:

Write and test your SQL queries using any SQL database system (e.g., MySQL, PostgreSQL, SQLite).

Ensure that your queries return the correct results and are optimized for performance.

- List all colleges with their deans:
- Find all departments within a specific college:
- Retrieve the details of the faculty member chairing a specific department:
- List all courses offered by a specific department:
- Find all sections of a specific course offered in a particular semester and year:
- List all students enrolled in a specific section:
- Calculate the average grade for a student across all sections they have completed:
- Find all instructors teaching a specific level of course (e.g., junior level):

Submission:

Submit a report that includes:

- The ER diagram.
- The normalization process and the resulting 3NF schemas.
- The relational schemas with primary keys and foreign keys identified.
- The SQL queries and their output results.

Evaluation Criteria:

- ER Diagram (30%): Completeness, correctness, and clarity.
- Normalization (20%): Correct identification of functional dependencies and normalization process.
- Relational Schemas (20%): Correct conversion from ER diagram, appropriate use of primary and foreign keys.
- SQL Queries (30%): Correctness, efficiency, and completeness of the queries.

Submission Deadline

Complete the following Part A, these questions by June 5, 2024 and upload the files to NLearn:

- ❖ **Question 01 – ER Diagram**
- ❖ **Question 02 – Relation Schema**
- ❖ **Question 03 – Normalization**

Complete Part B, Question 04 – SQL Queries section by June 26, 2024, and upload the files to Nlearn.